

challenge 3: Forklift with  
manipulator

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## PDR (Gruppe 2)



# Intro

## Robot design challenge 3: Forklift with manipulator

A forklift with a robot arm for lifting goods and pallets.

- Robot arm requirements:
  - a) Minimum 3-dof arm for lifting
  - b) Additional dof in gripper/fork to position fingers horizontally
  - c) Minimum reach with stationary mobile base is 150 cm in the horizontal plane, and 300 cm in the vertical plane
  - d) Capable of lifting:
    - a) Gripper: 100kg
    - b) Payload (picked object): Max. 500kg
- Mobile base requirements:
  - a) Capable of navigating autonomously between aisles in a 100 sqm warehouse
  - b) Have a turn radius less than 200 cm
  - c) Rear wheels are used for steering
- Sensor requirements:
  - a) Low-cost 3D camera for manipulation, type RealSense D435 or similar – mounted on mobile base
  - b) Localization sensors for autonomous navigation
- Expected results: A forklift capable of picking up a pallet at location A, navigating to location B, and placing the pallet correctly on a storage shelf.



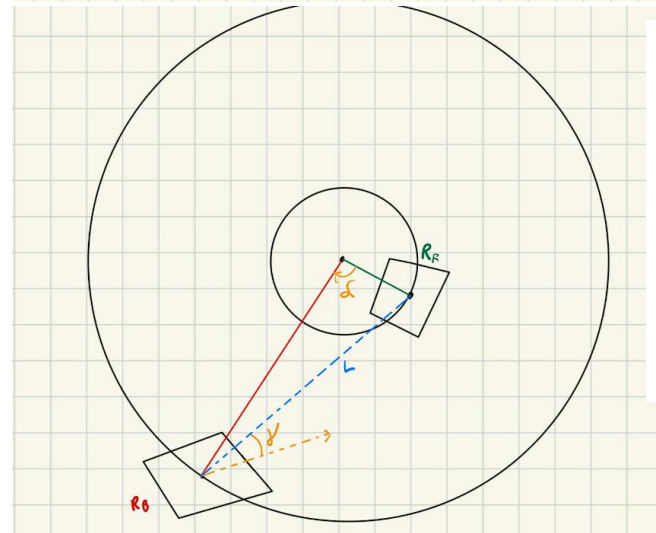
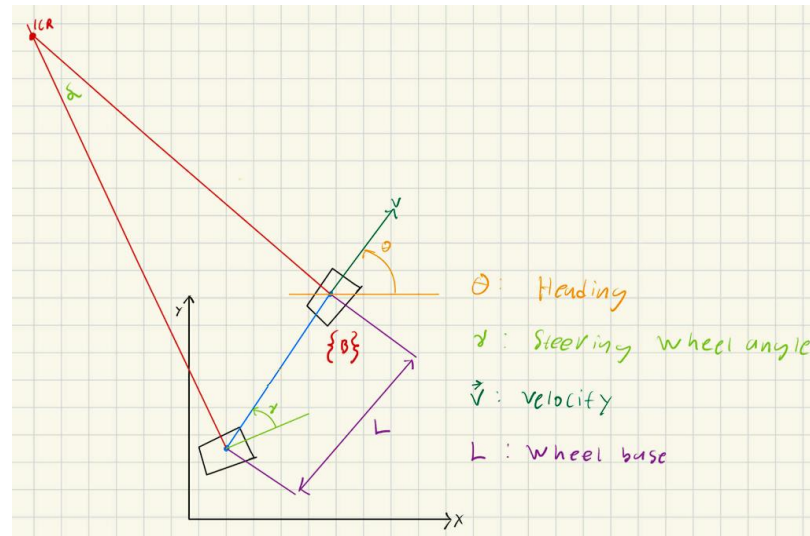
The Mini Excavator from the Toyota Challenge can be used for inspiration.



Design from ELE306 group 5, autumn 2022 (Bunes, Eide, Fimreite, Helland Myhren). For illustration only.

# Bevegelses mønster

- Lattice planner (Navigation)
- Pose regulator/Moving to a pose (Control)
- Localization and mapping/SLAM (Localization)
- Sensors ( Gyro, akselerometer, magnetometer, 3D kamera)



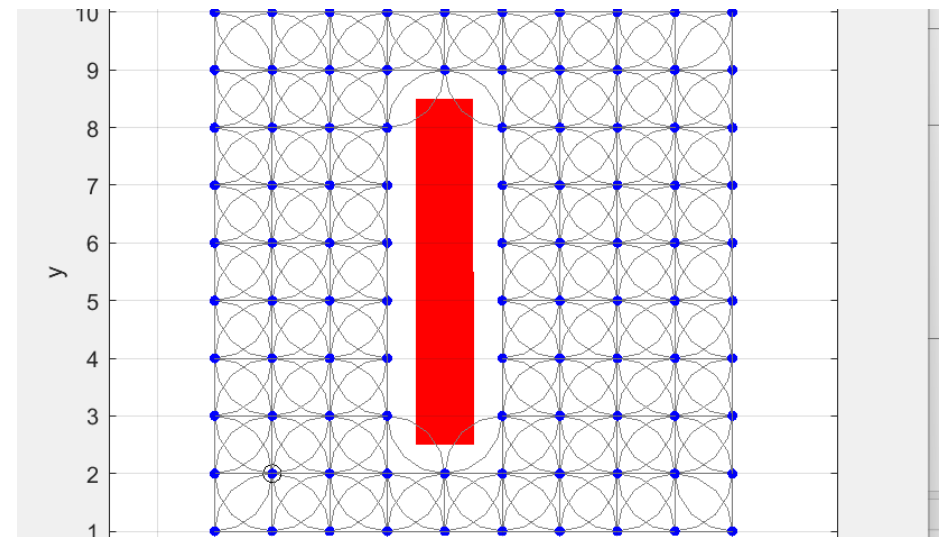
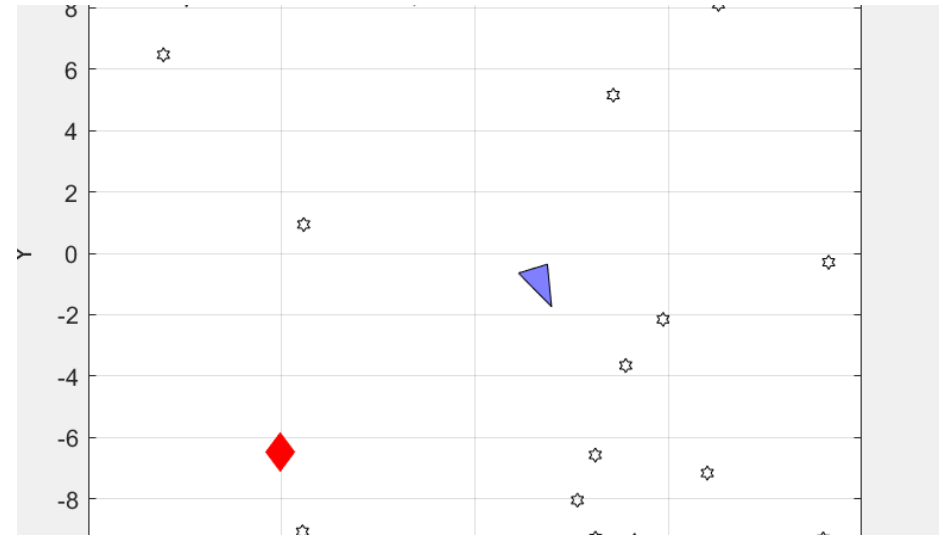
$$\dot{x} = v \cos \theta$$

$$\dot{y} = v \sin \theta$$

$$\dot{\theta} = \frac{v}{R_f} = \frac{v}{L} \tan \gamma$$

# Matlab scripts til simulering

- m\_lattice road
- m\_localization\_SLAM



## Robotikk prosjekt plan

[illegible]